Sperm capacitation test

There is insufficient evidence supporting the predictive value or clinical utility of this test. Additional quality studies are needed to support improved clinical outcomes.

Sperm capacitation is the set of natural physical changes that a spermatozoon undergoes to be able to fertilize the ovum. This occurs in vivo following ejaculation when the spermatozoa come into contact with the different fluids in the female genital tract.

The Cap-Score™ Sperm Function Test (Androvia LifeSciences, Mountainside, NJ), is an in vitro, laboratory-developed test designed to evaluate sperm function, particularly regarding capacitation. This assay identifies and analyzes the localization patterns of the ganglioside GM1 to evaluate the fertilizing ability of sperm. Conducting a Cap-Score test involves the incubation of sperm in medium containing capacitating stimuli (Cap) and non-capacitating (non-Cap) medium. The sperm that react to the capacitation stimuli are identified by specific GM1 localization patterns. The final data, called the "Cap-Score" reports the proportion of sperm within a sample that display the localization patterns that correspond with capacitation (Moody, 2017)

Sperm DNA integrity/fragmentation tests

There is insufficient evidence supporting the predictive value or clinical utility of this test. Prospective studies directly evaluating the impact of DNA fragmentation testing on the management of infertility are needed.

According to ASRM: "For a diagnostic test to be clinically useful the results must be reproducible, applicable to a given patient, and change the management of the patient. For tests of DNA integrity to be clinically important there must be an association of sperm DNA damage with reproductive outcomes" (ASRM, 2013).

At the current time, there is a lack of studies demonstrating that sperm DNA fragmentation testing results in improved clinical outcomes (improves the likelihood of conception).

Sperm penetration assays

There is insufficient evidence supporting the clinical utility of this test in lieu of newer technologies for treating male infertility.

ASRM contributed the following recommendation to Choosing Wisely®:

Don't perform advanced sperm function testing, such as sperm penetration or hemizona assays, in the initial evaluation of the infertile couple. Studies document that extreme variability exists among these tests, with very little correlation between results and outcomes. They have also been shown not to be cost-effective and often lead to more expensive treatments (ASRM, 2019).